Technical Specifications for BIRDS (Border Incident Reporting & Documentation System) - Phase II

1. **Basis of Evaluation**: Demo/Documentation must be submitted. The submitted demo/Documentation must be qualified/accepted by the Technical Evaluation Committee (TEC) as per the Technical Specifications for the bidder to be considered responsive.

2. The tenderer will be considered Non-Qualified or Non-Responsive if the required technical documents and the application demo/Documentation are not submitted or accepted by the TSC.

3. Any clarification must be submitted to TSC within 3 days (72 hours) from the tender opening date and time. The acceptance test of the application will be conducted on a No Cost, No Obligation basis. All necessary test results, reports (if any), and relevant certificates must be submitted along with the offer.

4. The tenderer will be considered NON-QUALIFIED or NON-RESPONSIVE if any modifications are made in the Technical Specification column. The tenderer should indicate compliance by stating "Comply" or "Not Comply" as applicable.

5. The related services shall comply with following technical specifications:

Ser	Item Description	Technical Specification	Bidder's Response
1.	System Type	Enterprise-grade Border Incident Reporting & Documentation System (BIRDS) for centralized recording, management, verification, tracking, analysis and reporting of border-related incidents.	
2.	Deployment Architecture	Centralized web-based application with secure access through BGB's authorized network/intranet. The system shall support scalable 3-tier architecture comprising presentation, application and database layers.	
3.	User Management	Role-Based Access Control (RBAC) with configurable roles, privileges and access scopes. User access shall be assignable based on HQ, Sector, Battalion and other authorized organizational levels.	
4.	Authentication	Secure username/password authentication with configurable password policy. The system shall be capable of integration with Active Directory/LDAP or other centralized identity management systems.	
5.	Incident Registration	Authorized users shall be able to create, edit, review and submit incident reports through standardized electronic forms.	
6.	Incident Classification	Configurable incident categories, sub-categories, incident types, severity levels and status classifications.	
7.	Incident Details	The system shall capture incident date/time, location, border area, BOP/Unit, incident type, description, persons/entities involved, response/action taken, outcome and other relevant information.	
8.	Geographical Information	Support for GPS coordinates (latitude/longitude) and map-based incident visualization. The system shall allow authorized users to identify and view incidents geographically.	
9.	Border Sector Information	Incident records shall be linked to relevant Region/Sector/Battalion/BOP/area of responsibility through configurable organizational hierarchy.	
10.	Evidence & Document Management	Support attachment and storage of photographs, videos, scanned documents, PDFs, reports and other evidence files against individual incidents.	

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11.	Media Metadata	The system should preserve available metadata of uploaded evidence, including file name, date/time and other relevant metadata where technically supported.	
12.	Incident Timeline	Each incident shall maintain a chronological history of reporting, verification, investigation, updates, actions and closure.	
13.	Workflow Management	Configurable workflow for Draft → Submitted → Reviewed → Verified/Approved → Closed, with authority-based actions at each stage.	
14.	Verification & Approval	Authorized supervisory officers shall be able to review, return, modify, verify, approve or reject incident reports.	
15.	Incident Tracking	Unique system-generated incident/reference number for every registered incident. Users shall be able to track incidents by reference number, status, location, date, category and responsible unit.	
16.	Search Facility	Advanced search and filtering by date range, location, unit, incident type, severity, status, reference number and other configurable parameters.	
17.	Dashboard	Role-based dashboard displaying incident statistics, pending reports, verified incidents, open/closed cases, trends and other operational indicators.	
18.	GIS/Map Dashboard	Map-based visualization of incidents with configurable markers, filters and incident categories.	
19.	Analytics & Statistics	Generation of statistical and analytical reports based on incident type, location, time period, unit, frequency, severity and other parameters.	
20.	Trend Analysis	Support for graphical presentation of incident trends by day, week, month, quarter and year.	
21.	Reporting	Generation of configurable operational, statistical and management reports in printable and downloadable formats.	
22.	Export	Authorized users shall be able to export permitted reports/data in formats such as PDF, Excel/CSV.	
23.	Notifications	Configurable system notifications for new submissions, pending verification, returned reports, approvals, escalations and other workflow events.	
24.	Audit Trail	Comprehensive and tamper-evident audit logging of user login, creation, modification, approval, deletion, download and other significant system activities.	
25.	Data Integrity	The system shall maintain data consistency and shall prevent unauthorized alteration or deletion of finalized/approved incident records.	
26.	Data Security	Sensitive information shall be protected through appropriate access controls, secure application architecture and encryption mechanisms.	
27.	Encryption	Communication between client and server shall use HTTPS/TLS. Sensitive data shall be encrypted at rest where applicable.	
28.	Session Security	Configurable session timeout, secure session management, concurrent-session control and automatic logout after defined inactivity period.	
29.	Password Security	Configurable minimum password length, complexity, failed-login threshold, password expiry and account-lockout policy.	
30.	Database	Enterprise-grade relational database supporting transactional integrity, indexing, backup, recovery and scalable data storage.	
31.	API / Integration	Secure REST API and/or equivalent integration interface for interoperability with authorized BGB systems and future applications.	

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32.	AD/LDAP Integration	Capability to integrate with existing BGB Active Directory/LDAP infrastructure for centralized authentication and user management.	
33.	File Storage	Centralized and scalable storage mechanism for incident-related documents and multimedia evidence with configurable storage limits and retention policies.	
34.	Backup	Automated scheduled backup of application/database data and associated documents/evidence. Backup frequency shall be configurable.	
35.	Disaster Recovery	The system shall support deployment/replication at an authorized Disaster Recovery (DR) site and restoration of application and database services following system failure.	
36.	High Availability	Architecture shall support redundancy/high availability of critical application and database components where required by the deployment environment.	
37.	Scalability	The system shall be scalable to accommodate increasing numbers of users, incidents, attachments, organizational units and historical records without major architectural redesign.	
38.	Concurrent Users	The proposed system shall support a minimum of [100/250/500] concurrent users without degradation of core system functions.	
39.	Performance	Normal user transactions shall generally respond within 3 seconds under the specified concurrent-user and infrastructure load, excluding large file uploads/downloads and complex report generation.	
40.	Availability	The application shall support a minimum 99.5% service availability, excluding scheduled maintenance and force majeure events.	
41.	Mobile Responsiveness	Web interface shall be responsive and usable on authorized desktop, laptop and tablet devices.	
42.	Browser Support	Compatible with current supported versions of Google Chrome, Microsoft Edge and Mozilla Firefox.	
43.	Localization	Interface shall support English and shall be capable of supporting Bangla labels/content where required.	
44.	Data Retention	Configurable data-retention policy with controlled archival of historical incident records.	
45.	Data Archiving	Authorized administrators shall be able to archive historical records while retaining their audit trail and retrievability.	
46.	Administrative Module	Central administration module for managing users, roles, organizational units, incident categories, workflows, system parameters and reference data.	
47.	Configuration	Incident types, categories, workflow stages, organizational hierarchy, notification rules and report parameters shall be configurable without modification of core application code wherever practicable.	
48.	Logging & Monitoring	Application, security and system logs shall be maintained with configurable retention and administrative monitoring facilities.	
49.	Security Hardening	Application shall follow recognized secure development practices and shall address common web application vulnerabilities including those identified in the OWASP Top 10.	
50.	Vulnerability Assessment	Supplier shall conduct application security testing/Vulnerability Assessment and provide a report before production deployment. Critical and high-risk vulnerabilities shall be remediated before acceptance.	

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51.	Source Code / Ownership	All application components developed specifically for BGB shall be delivered as per tender/contract terms, including relevant source code, database schema, configuration and technical documentation where specified.	
52.	Documentation	Supplier shall provide System Administration Manual, User Manual, Installation/Deployment Guide, API Documentation, Database Documentation and Backup/Recovery Procedure.	
53.	Training	Vendor shall provide hands-on training for system administrators, designated operators and end users, including incident entry, verification, reporting and administration.	
54.	Warranty & Support	Minimum 01 (One) year warranty and technical support including bug fixing, security patches, configuration support and troubleshooting.	
55.	Customization	System shall allow reasonable customization of forms, reports, workflows, organizational hierarchy and incident classifications according to BGB operational requirements.	
56.	Deployment	Complete installation, configuration, integration, testing, data migration (if applicable), commissioning and production deployment shall be included in the scope of supply.	
57.	Acceptance Testing	Final acceptance shall be based on successful completion of Functional Acceptance Test (UAT) against the approved requirements and test cases.	
58.	Future Expansion	The architecture shall allow future integration with GIS, mobile applications, surveillance systems, digital evidence systems and other authorized BGB information systems.	

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